TERMOJET

SERVICE MANUAL

Variable-frequency in-line circulating pump

Model: **TBE**

WARNING

- Before operation, make sure that the electric pump is reliably earthed and a residual current device is fitted.
- Do not touch the electric pump while it is running.
- Do not run the electric pump without water — dry running destroys the seals.

TERMOJET · boiler room equipment

termojet.com.ua

Contents

Thanks for choosing our product. Read these service instructions carefully before installation and use, and keep this manual in your safekeeping for the whole service life of the pump.

Warnings

Prohibited personnel warning

1. Any child or any adult who has any physical, sensory or mental defects, or who lacks the relevant experience or knowledge, may use this product only if supervised or given the method of safe use of this product as well as knowing the dangers involved.
2. No child shall play with this product as a toy.
3. Without supervision, no child shall be allowed to clean or maintain this product.

Pressure warning

1. The system where the pump lies shall be able to withstand the maximum pressure of the pump.

Electricity warning

1. The electric power system may be used only when it has the safety protection measures specified by the existing provisions of the country where the product is installed.

Modification-related warning

1. Where any electric pump is tampered with, modified and/or operated outside the recommended operating scope, or goes against any other instruction given in this manual, the manufacturer will not guarantee the correct operation of the electric pump and will not be responsible for any loss which might be caused by the electric pump.
2. The manufacturer refuses to undertake any responsibility for any error which might appear in this manual due to misprint or misreplication. The manufacturer reserves the right to make any modification to the product which, in its opinion, is necessary or useful, without affecting the basic features of the product.

I. Symbols in the manual

Warning. Failure to follow the safety instructions may cause personal injury.

Careful. Failure to follow the safety instructions may result in equipment failure.

Remark. Reasons for operation.

II. Overview

The TBE variable-frequency pipeline pump is a product that achieves variable-frequency, constant-pressure control with the installation of components such as a frequency converter and a pressure tank, based on the TB pump.

The series has many advantages — compact structure, energy conservation and environmental protection, safety and reliability. The products are easy to disassemble and assemble, and are connected to the pipeline directly.

III. Identification

1. Pump appearance

2. Structure

The product is detachable at the top, with the water pump maintained without the plumbing being dismantled. The water pump is equipped with an IE3 series motor, a universal mechanical seal and a high-temperature resistant rubber sealing ring.

3. Model

TBE 50 – 35 / 2 – S

TBE pump category: variable-frequency in-line pipeline pump

50 diameter of pump inlet and outlet, mm — DN 50

35 prescribed lift, m

2 number of motor poles

S material of the impeller: stainless steel (no index — cast iron)

IV. Delivery and lifting

1. Delivery

The water pump has been strictly tested and inspected before leaving the factory and specially packaged to be handled, loaded and unloaded by forklifts or similar equipment.

2. Lifting

WARNING
All motors are equipped with lifting rings to lift the entire pump, which shall be lifted by two lifting rings simultaneously rather than by one single lifting ring.

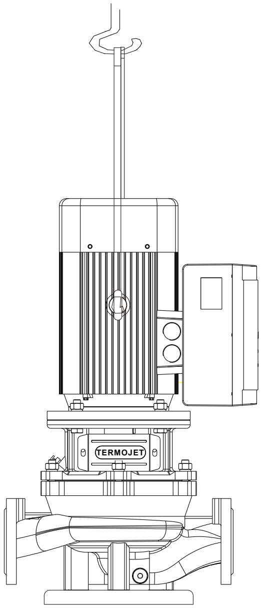


Figure 1. Lifting the water pump by nylon straps

V. Application

1. Working conditions

1. Working pressure ≤ 1.6 MPa; determined according to order requirements in a special environment.
2. The ambient temperature shall not exceed 40 °C and the relative humidity shall not exceed 95 %.
3. The pH value of the conveying medium is 5–9, and the medium temperature is 0 °C to +120 °C.
4. The solid volume ratio of the conveying medium is ≤ 0.2 %, and the solid particles are ≤ 0.2 mm.
5. The pump shall be kept away from flammable and explosive objects and heat sources, and installed on flame-retardant materials such as metals.
6. If the pump is installed in an electrical cabinet or another enclosed object, it is necessary to install fans or other cooling equipment inside the cabinet and to provide ventilation openings, so that the ambient temperature stays below 40 °C; otherwise the frequency converter may be damaged by the excessive ambient temperature.

2. Application site

TB series water pumps are used in cold and hot water transportation, pressurization and circulation systems:

- heating system;
- regional water supply system;
- secondary pressurization of high-rise pipeline network;
- boiler feed-water;
- industrial cleaning system;
- ordinary industrial water systems;
- air conditioning circulation and cooling system;
- general domestic water.

> The operating point must be within the specified performance range of the water pump — only then will the pump operate safely and efficiently.

3. Liquid conveyed

The liquid conveyed shall be clean, of low viscosity, non-explosive, and free from solid particles and fibrous substances that mechanically or chemically damage the water pump:

- water for a heating system (the water quality shall meet the standard requirements for a hot water supply system);
- cooling fluid;
- industrial liquid;
- softened water;
- residential hot water.

The following may occur if the density and viscosity of the liquid conveyed are greater than those of ordinary clean water: a significant decrease in pressure, significantly low hydraulic performance, a significant increase in motor energy consumption. In this case the water pump must be equipped with a higher power motor — please contact your Termojet supplier for the specific selection.

It is necessary to select the O-ring seals according to the situation when the liquid conveyed contains minerals, oil substances, chemicals or any other liquid that is different from clean water.

VI. Installation

1. Installation environment

> **WARNING**
> It is necessary to take protective measures to ensure the personal safety of personnel if a high-temperature, low-temperature, toxic or corrosive liquid is transported.

The water pump must be installed in a dry, cool, well ventilated place with an ambient temperature not lower than 0 °C (frost-free).

It is necessary to confirm the direction of flow by the arrow on the pump casing before connecting the pump to the pipeline. To ensure the sealing of the pipeline connection, install gaskets made of rubber or another sealing material between the flanges, as shown in Figure 2.

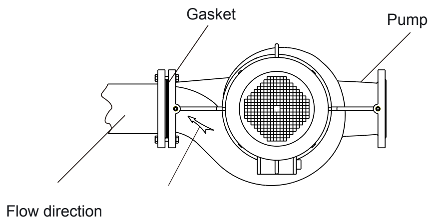


Figure 2. Connection to the pipeline

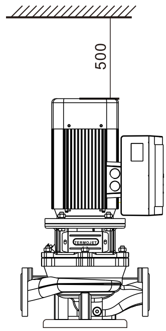


Figure 3. Clearance for maintenance

It is necessary to leave sufficient space around the pump to ensure normal air circulation and subsequent disassembly and maintenance, as shown in Figure 3.

2. Pipeline connection

- Install an isolation valve at the inlet and the outlet of the pump to prevent draining the system during maintenance.
- Connect the water pump directly to the pipes if the pipes on both sides of the pump can withstand its weight. This applies to water pumps with a motor power of up to 2.2 kW inclusive.
- Install water pumps with a motor power of 3 kW and above on a concrete foundation, and install a shock absorber if possible. Of course, this is also applicable to pumps below 3 kW. See subsection 3 for details.
- It is important to ensure that the water pump is not pulled by the tension of the pipeline when the pipeline is installed.
- It is necessary to fully consider the suction performance of the pump and the operating pressure of the pipeline, so a reasonable diameter of the inlet and outlet pipes must be designed before installation.
- Do not install the water pump at the lowest point of the system; otherwise there will be dirt deposition.
- Avoid air pockets during the installation of the pipeline, especially at the inlet end of the pump, as shown in Figure 5.

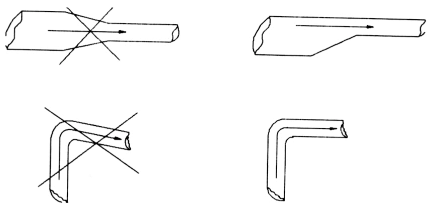


Figure 5. Correct installation of the pipeline on the suction side of the water pump

3. Noise reduction and vibration reduction

The vibration and noise of equipment are inevitable, but it is possible to take effective measures to greatly reduce them while ensuring the safe operation of the equipment. Based on the vibration reduction experience in the industry, we recommend the following:

- it is necessary to consider installing a shock absorber for a water pump with $2.2 < P \leq 7.5$ kW;
- a water pump with $11 \leq P \leq 18.5$ kW must be equipped with a concrete foundation.

Noise and vibration are caused by various factors, such as the operation of the motor and the pump, as well as water flow in the pipeline and valves. Currently, the most widely used method is a concrete foundation with shock absorbers and expansion soft joints, as shown in Figure 6.

1	Concrete foundation
2	Shock absorber
3	Inlet valve
4	Expansion soft joint
5	Straight pipe
6	Outlet valve

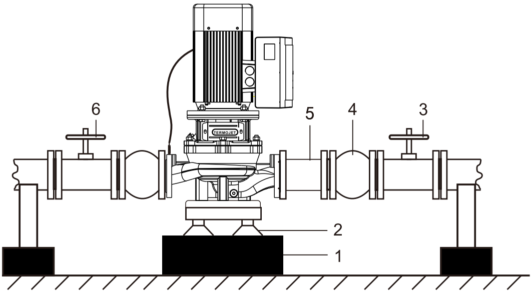
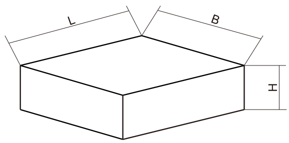


Figure 6. Vibration reduction of the water pump

As the simplest and most practical measure for vibration reduction, the concrete foundation is sturdy enough to provide permanent support for the entire pump, absorbing normal stress, impact and vibration. The following dimensions are recommended for the concrete foundation:

Weight of pump G, kg	Length L, mm	Width B, mm	Height H, mm
$150 \leq G < 200$	620	620	300
$200 \leq G < 300$	720	720	350
$300 \leq G < 400$	800	800	400
$400 \leq G < 500$	850	850	425
$500 \leq G < 600$	900	900	450
$600 \leq G < 700$	950	950	475
$700 \leq G < 800$	1000	1000	500
$800 \leq G < 900$	1050	1050	525
$900 \leq G < 1000$	1050	1050	550
$1000 \leq G < 1100$	1100	1100	550
$1100 \leq G < 1200$	1150	1150	560
$1200 \leq G < 1300$	1150	1150	580
$1300 \leq G < 1400$	1200	1200	600
$1400 \leq G < 1500$	1200	1200	610
$1500 \leq G < 1600$	1250	1250	620



VII. Startup and maintenance

1. Before startup

CAREFUL
Do not start the water pump until the pump body is filled with liquid and the air is completely discharged.

WARNING
Pay attention to the direction of the exhaust hole during venting, and make sure that the liquid discharged (especially high-temperature, low-temperature, toxic and corrosive liquid) will not harm personnel or the motor and other components.

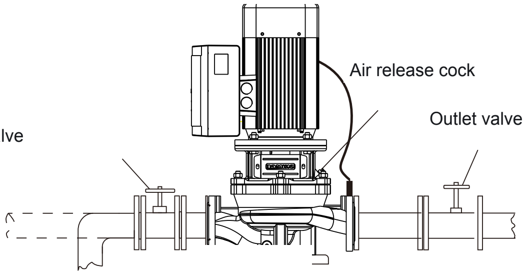


Figure 8. Valves for filling and venting

Closed or open pipeline system with liquid level lower than the pump inlet

1. Close the outlet valve and open the inlet valve.
2. Unscrew the vent cock on the pump cover.
3. Unscrew the plug at the upper end of the flange on either side of the water pump, and confirm the specific position according to the water pump.
4. Fill the entire pump body and suction pipeline through the water injection hole.
5. Tighten the plug again.
6. Tighten the vent cock as shown in Figure 8.

Closed or open pipeline system with liquid level higher than the pump inlet

1. Close the inlet valve and unscrew the vent cock on the pump cover, as shown in Figure 8.
2. Slowly open the inlet valve until the liquid evenly flows out of the exhaust hole.
3. Tighten the vent cock and fully open the valve.

2. Rotation inspection

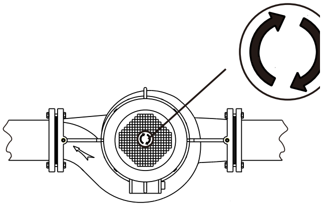


Figure 9. Direction of rotation by the arrow on the casing

> **CAREFUL**
> Do not start the water pump to check the rotation direction of the motor until the pump body is filled with liquid and the air is completely discharged.

3. Start

1. Make sure that the inlet valve is fully open and the outlet valve is slightly open before starting the water pump.
2. Start the water pump.
3. After starting, adjust the valve to bring the flow to the required working condition, checking the pressure gauge, the flow meter and the ammeter.
4. Assign a dedicated person to inspect the water pump regularly when it operates normally, and to shut the pump down for inspection and maintenance in a timely manner in case of abnormalities. Otherwise the water pump will be permanently damaged.

4. Maintenance

Pump

Inspect and maintain the water pump regularly. Check the wear of the mechanical seal regularly if the pumped medium contains complex substances. Inject an appropriate amount of silicone oil between the shaft and the shaft seal to prevent the seal from being corroded and bonded if the pump is not used for a long time.

Inspect the motor regularly and keep it clean and ventilated; clean and inspect the motor more often if the water pump is installed in a dusty environment.

Frequency converter

WARNING

- Do not touch the terminals of the frequency converter when it is powered on — this may cause electric shock.
- Designate qualified electrical engineering personnel for maintenance, inspection or component replacement.
- Start work at least 10 minutes after the converter is powered off, or after confirming that there is no residual voltage; otherwise personal injury may occur.
- Do not touch the board with your hands: there are CMOS integrated circuits on the PCB, which static electricity may damage.
- It is strictly prohibited to modify the frequency converter without permission; otherwise it may cause serious personal injury or death.

VIII. Operation panel of the frequency converter

1. Schematic diagram

1. **MENU** — used to go from the display mode to the parameter mode.
2. **P.SP/ENT** — the shortcut key for setting the water pressure, and the confirmation key that confirms and saves a parameter value.
3. **SHIFT** — used to move the cursor when switching the display content and modifying parameters. Hold SHIFT in the shutdown state to run the pump at 5 Hz and check whether the direction of rotation is right or wrong. During operation, SHIFT switches between the operating frequency, the output current, the set pressure and the feedback pressure. The flashing bit is the bit currently being modified.
4. ▲ ▼ — used to modify parameter values and the set pressure. Holding them for 2 seconds enters the pressure setting mode.
5. **RUN/STOP** — starts the pump; the same key stops the pump and resets a fault.

2. Indicator light

- **Operation** — always on: the pump is running; flashing: sleep shutdown indication.
- **Stop** — stop indication (standby indication).
- **Net pump** — always on when the multi-pump network is successfully connected, and off when it fails to be connected.
- **Alarm** — frequency converter protection alarm.
- **Set pressure** — the value on the display screen is the set pressure value.
- **Feedback pressure** — the value on the display screen is the feedback pressure value of the pipeline network.

3. Key operation

The menu has three levels:

1. function code group number (first-level menu);
2. function code label (second-level menu);
3. function code setting value (third-level menu).

Press **MENU** or **P.SP/ENT** in the third-level menu to return to the second-level menu. The difference: **P.SP/ENT** saves the set parameter on the control board, returns to the second-level menu and automatically transfers to the next function code label; **MENU** returns directly to the second-level menu without storing the parameter, and remains at the current function code label.

If the parameter does not flash in the third-level menu state, it means that the bit cannot be modified — press the **SHIFT** key to switch to the flashing bit.

> The parameters marked with « ● » in the parameter table are modified in the shutdown state only. The parameters marked with « ◎ » are the actual detection values and cannot be changed.

For example: change F0.02 from 0 to 1

0.00 → MENU → F0 → P.SP/ENT → F0.00 → ▲ ▼ → F0.02 → P.SP/ENT → 0
→ ▲ → 1 → P.SP/ENT → F0.03 → MENU → F0 → MENU → 0.00

- In the stopped or running state the first four bits flash (**0.00**); in the group and code menus the first bit flashes.
- If a value has many bits, press ▲ and ▼ at the same time to move to the next bit.
- The ▲ ▼ keys select other parameters of the F0 group, and other parameter groups in the first-level menu.
- Hold the **MENU** key for 2 seconds to enter a parameter group.
- The final state **0.00** is the stopped or running state (frequency 50 Hz).

4. Pressure setting

Hold the ▲ and ▼ keys for 2 seconds in the running or shutdown state to enter the pressure setting mode. Use the ▲ ▼ keys to set the target pressure and exit with **MENU** or **P.SP/ENT** — the pressure value is saved automatically.

> Pressure conversion equation: 0.1 MPa = 100 kPa = 1 bar = 1 kgf/cm².

IX. Technical data

1. Model parameters

No.	Model	Prescribed flow rate (m^3/h)	Prescribed lift (m)	Maximum pressure setting (m)	Displayed pressure setting value (MPa)	Motor power (kW)
1	TBE32-18/2	12,5	18	18	0,18	1,1
2	TBE32-21/2	12,5	21	21	0,21	1,5
3	TBE32-26/2	12,5	26	26	0,26	2,2
4	TBE32-33/2	12,5	33	33	0,33	3
5	TBE32-40/2	12,5	40	40	0,40	3
6	TBE32-50/2	12,5	50	50	0,50	4
7	TBE40-16/2	12,5	16	16	0,16	1,1
8	TBE40-21/2	12,5	21	21	0,21	1,5
9	TBE40-20/2	20	20	20	0,20	2,2
10	TBE40-26/2	20	26	26	0,26	3
11	TBE40-30/2	25	30	30	0,30	4
12	TBE40-36/2	25	36	36	0,36	5,5
13	TBE40-50/2	25	50	50	0,50	7,5
14	TBE50-12/2	16	12	12	0,12	1,1
15	TBE50-15/2	20	15	15	0,15	1,5
16	TBE50-18/2	25	18	18	0,18	2,2
17	TBE50-24/2	25	24	24	0,24	3
18	TBE50-28/2	30	28	28	0,28	4
19	TBE50-36/2	30	36	36	0,36	5,5
20	TBE50-40/2	35	40	40	0,40	7,5
21	TBE50-50/2	40	50	50	0,50	11
22	TBE50-60/2	50	60	60	0,60	15
23	TBE50-71/2	50	71	71	0,71	18,5
24	TBE65-12/2	25	12	12	0,12	1,5
25	TBE65-15/2	30	15	15	0,15	2,2
26	TBE65-20/2	30	20	20	0,20	3
27	TBE65-22/2	40	22	22	0,22	4

No.	Model	Prescribed flow rate (m^3/h)	Prescribed lift (m)	Maximum pressure setting (m)	Displayed pressure setting value (MPa)	Motor power (kW)
28	TBE65-30/2	40	30	30	0,30	5,5
29	TBE65-34/2	50	34	34	0,34	7,5
30	TBE65-42/2	50	42	42	0,42	11
31	TBE65-52/2	50	52	52	0,52	15
32	TBE65-60/2	60	60	60	0,60	18,5
33	TBE80-13/2	50	13	13	0,13	3
34	TBE80-19/2	50	19	19	0,19	4
35	TBE80-23/2	50	23	23	0,23	5,5
36	TBE80-29/2	50	29	29	0,29	7,5
37	TBE80-30/2	50	30	30	0,30	11
38	TBE80-38/2	50	38	38	0,38	15
39	TBE80-47/2	80	47	47	0,47	18,5
40	TBE100-10/2	80	10	10	0,10	3
41	TBE100-15/2	80	15	15	0,15	4
42	TBE100-17/2	60	17	17	0,17	5,5
43	TBE100-22/2	80	22	22	0,22	7,5
44	TBE100-27/2	100	27	27	0,27	11
45	TBE100-34/2	100	34	34	0,34	15
46	TBE100-40/2	110	40	40	0,40	18,5
47	TBE125-11/4	120	11	11	0,11	5,5
48	TBE125-15/4	120	15	15	0,15	7,5
49	TBE125-18/4	160	18	18	0,18	11
50	TBE125-22/4	160	22	22	0,22	15
51	TBE125-28/4	160	28	28	0,28	18,5
52	TBE150-13/4	200	13	13	0,13	11
53	TBE150-17/4	200	17	17	0,17	15
54	TBE150-22/4	200	22	22	0,22	18,5

2. Test pressure and inlet pressure

Test pressure — the test value with clean water at a room temperature of 20 °C.

Inlet pressure must be adjusted correctly to ensure that the water pump is in the optimal operating state and the noise level is controlled at the lowest level.

Cavitation will occur in the water pump if the pressure inside the pump is lower than the gasification pressure of the conveyed medium. The maximum suction head H (m), which ensures the minimum pressure at the pump inlet, can be calculated by the formula:

$$H = P_b \times 10.2 - NPSH - H_f - H_v - H_s$$

P_b atmospheric pressure, bar; for a closed system — the system pressure, bar;

NPSH net suction head, m; read at the corresponding maximum flow rate on the performance curve;

H_f pipeline loss at the inlet, m, corresponding to the value at maximum flow rate;

H_v gasification pressure of the liquid, m, at the working temperature;

H_s safety margin, m, with a minimum of 0.5 m.

The water pump can be installed and operated by suction if H is a positive value. If H is a negative value, the pump can only be operated normally when it is installed with a positive inlet head.

Generally this is not calculated. It is needed only when:

1. the liquid temperature is high;
2. the liquid flow rate exceeds the limit value;
3. there is a large suction head or a long inlet pipeline;
4. the system pressure is low;
5. the conditions for the inlet are poor.

3. Noise pressure level

Three-phase motor with load, 50 Hz.

2P motor (2900 r/min)

Motor power (kW)	Speed (r/min)	Noise dB(A)
1,1	2880	65
1,5	2895	71
2,2	2895	72
3	2895	73
4	2905	74
5,5	2930	78
7,5	2930	78
11	2945	82
15	2945	82
18,5	2940	82

4P motor (1450 r/min)

Motor power (kW)	Speed (r/min)	Noise dB(A)
5,5	1460	70
7,5	1460	70
11	1465	72
15	1465	72
18,5	1470	72

For a water pump, the noise is mainly generated by the motor; the noise produced by the pump itself is very small and almost negligible. Therefore the motor noise level is used to define the noise level of the pump. Variable frequency noise needs to be considered for a motor with a frequency converter.

4. Protection level

The motor protection level is **IP55**.

5. Model selection of vulnerable parts

Motor bearings

2P motor

Motor model	Front bearing	Rear bearing
1.1 kW	6204	6204
1.5 kW	6305	6205
2.2 kW	6305	6205
3 kW	6306	6206
4 kW	6306	6206
5.5 kW	6308	6308
7.5 kW	6308	6308
11 kW	7309	6309
15 kW	7309	6309
18.5 kW	7309	6309

4P motor

Motor model	Front bearing	Rear bearing
5.5 kW	6308	6308
7.5 kW	6308	6308
11 kW	7309	6309
15 kW	7309	6309
18.5 kW	7311	6311

The «front bearing» is the bearing at the extension end of the motor shaft.

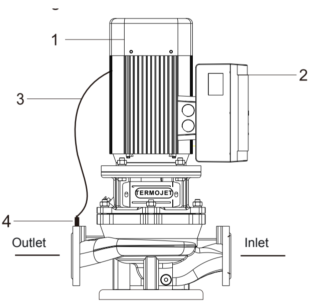
The following shall be noted for the use of bearings

- The motor should be filled or replaced with lubricating grease after it runs for about 2500 hours (sealed bearings do not need to be replaced during the service life).
- The grease should be replaced in a timely manner if the bearing overheats or the lubricating grease deteriorates during operation.
- The vibration and noise of the motor increase significantly at the end of the service life of the bearing. The bearing should be replaced when the radial clearance reaches the following values:

Inner diameter of bearing, mm	10–30	35–50	55–80	85–130
Ultimate radial clearance value, mm	0.10	0.15	0.20	0.30

- The brand of the replacement lubricating grease: ZL-3 lithium based grease or another similar grease.
- Grease-filling amount: for 2-pole motors fill ½ of the gap between the inner and outer rings of the bearing, for 4-pole motors ⅔.

6. Structural diagram

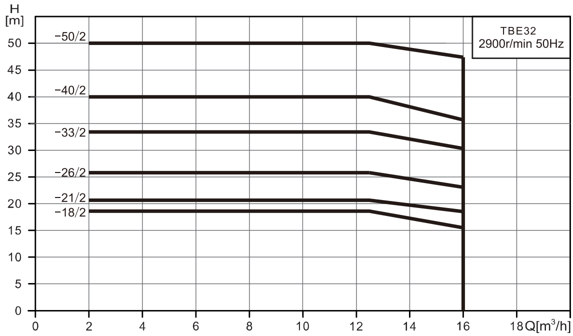


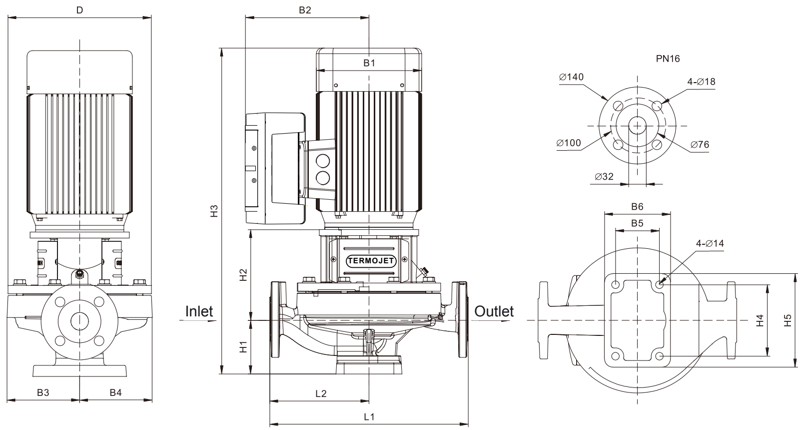
Main components:

1. TB water pump;
2. frequency converter;
3. data cable of sensor;
4. pressure sensor.

7. Installation dimensions and performance curves

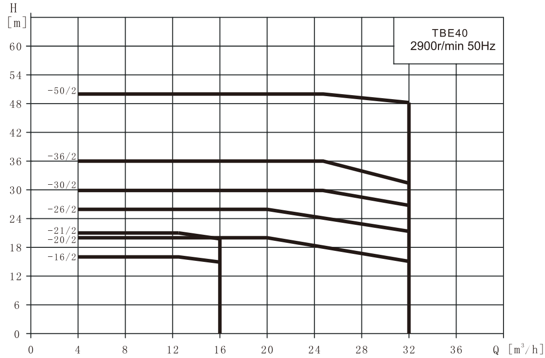
TBE32

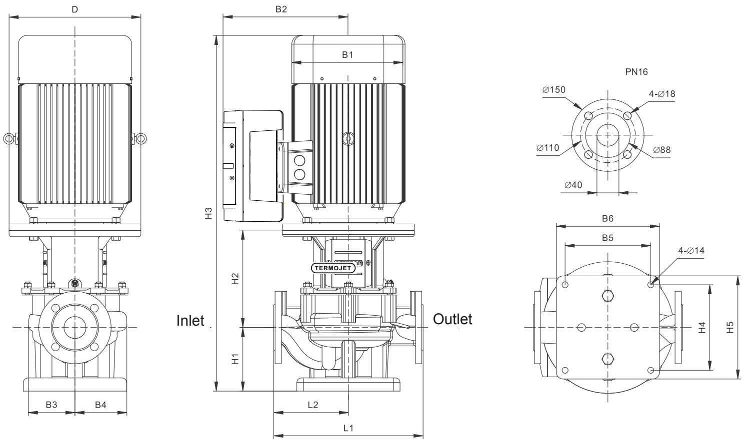




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE32-18/2	188	148	207	95	95	70	100	90	142	480	120	150	320	160	33
TBE32-21/2	188	166	226	95	95	70	100	90	149	520	120	150	320	160	37
TBE32-26/2	223	166	226	110	110	70	100	90	149	535	120	150	320	160	41
TBE32-33/2	223	191	239	110	110	70	100	90	159	560	120	150	320	160	59
TBE32-40/2	260	191	239	129	129	80	120	100	166	585	130	170	360	180	60
TBE32-50/2	260	212	251	129	129	80	120	100	166	615	130	170	360	180	63

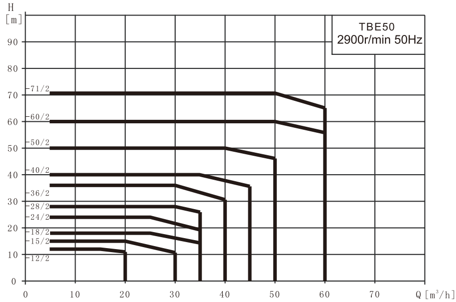
TBE40

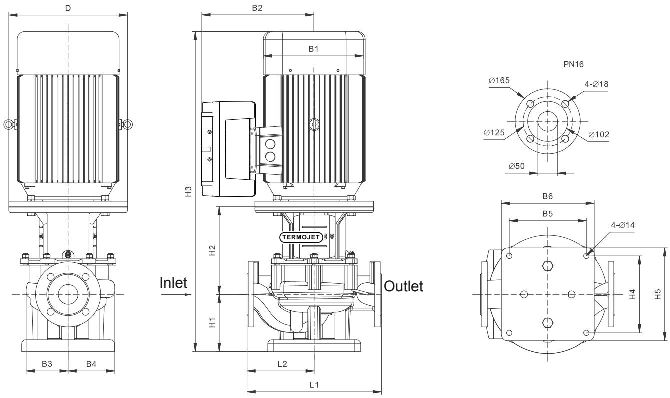




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE40-16/2	200	153	205	121	121	195	235	98	170	520	195	235	320	160	40
TBE40-21/2	200	168	217	121	121	195	235	98	170	561	195	235	320	160	45
TBE40-20/2	200	168	217	121	121	195	235	130	170	593	195	235	340	170	50
TBE40-26/2	250	195	232	121	121	195	235	130	190	637	195	235	340	170	61
TBE40-30/2	250	215	249	121	121	195	235	130	190	663	195	235	340	170	67
TBE40-36/2	300	260	271	167	167	195	235	140	225	785	195	235	440	220	98
TBE40-50/2	300	260	271	167	167	195	235	140	225	785	195	235	440	220	104

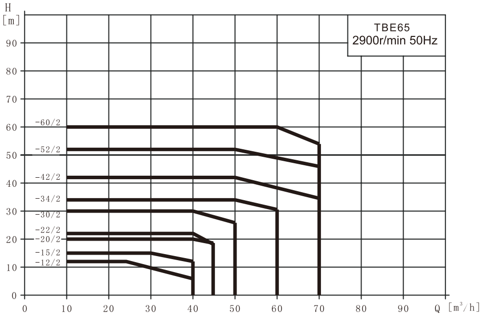
TBE50

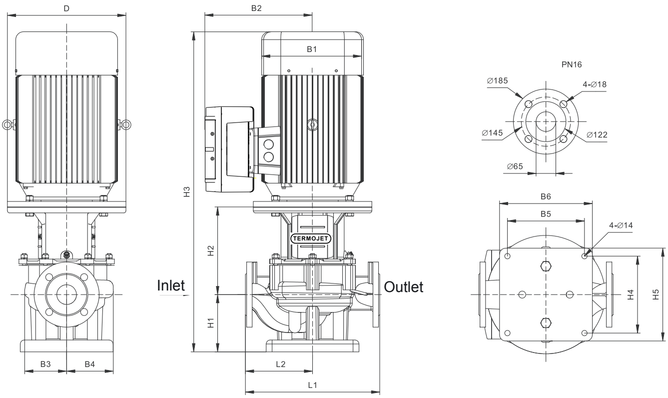




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE50-12/2	200	153	205	121	121	195	235	145	150	547	195	235	340	170	46
TBE50-15/2	200	168	217	121	121	195	235	145	150	588	195	235	340	170	50
TBE50-18/2	200	168	217	121	121	195	235	145	150	588	195	235	340	170	52
TBE50-24/2	250	195	232	121	121	195	235	145	170	632	195	235	340	170	63
TBE50-28/2	250	215	249	121	121	195	235	145	182	670	195	235	340	170	69
TBE50-36/2	300	260	271	121	121	195	235	145	222	787	195	235	340	170	90
TBE50-40/2	300	260	271	167	167	195	235	145	223	788	195	235	440	220	109
TBE50-50/2	350	314	382	167	167	195	235	145	258	901	195	235	440	220	172
TBE50-60/2	350	314	382	167	167	195	235	145	258	901	195	235	440	220	187
TBE50-71/2	350	314	382	167	167	195	235	145	258	945	195	235	440	220	206

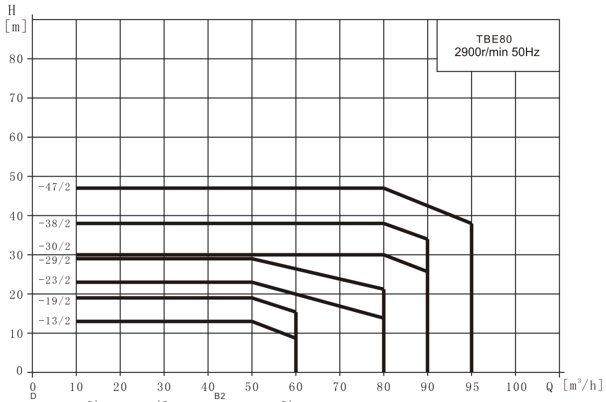
TBE65

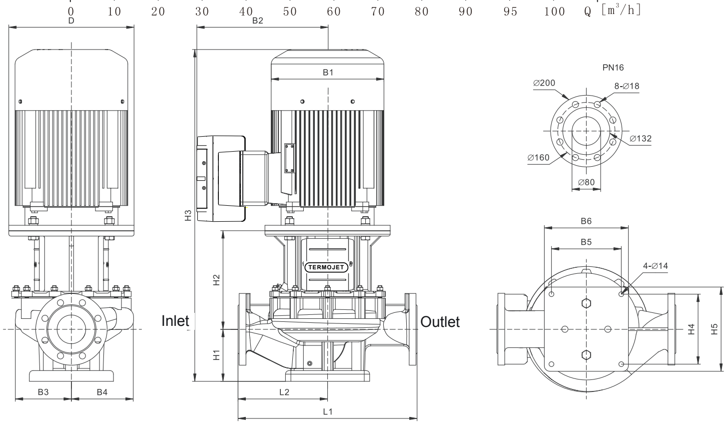




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE65-12/2	200	168	217	121	132	195	235	135	170	598	195	235	360	180	52
TBE65-15/2	200	168	217	121	132	195	235	135	170	598	195	235	360	180	54
TBE65-20/2	250	195	232	121	132	195	235	135	190	642	195	235	360	180	65
TBE65-22/2	250	215	249	121	132	195	235	135	190	668	195	235	360	180	71
TBE65-30/2	300	260	271	121	132	195	235	135	230	785	195	235	360	180	92
TBE65-34/2	300	260	271	121	132	195	235	135	230	785	195	235	360	180	98
TBE65-42/2	350	314	382	167	169	195	235	155	260	913	195	235	475	237,5	174
TBE65-52/2	350	314	382	167	169	195	235	155	260	913	195	235	475	237,5	190
TBE65-60/2	350	314	382	167	169	195	235	155	260	957	195	235	475	237,5	209

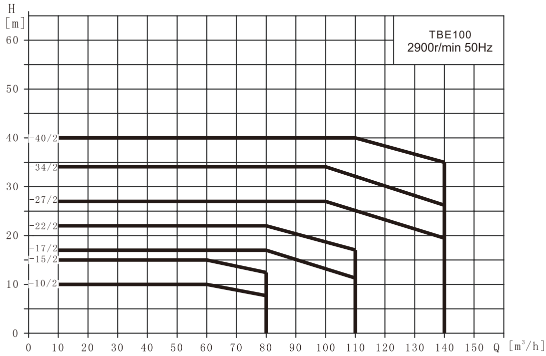
TBE80

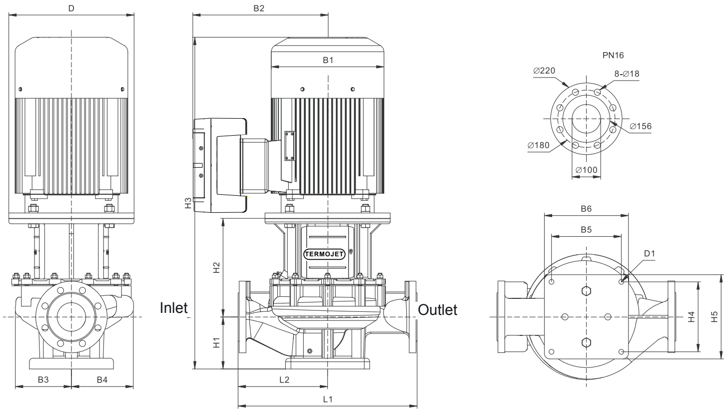




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE80-13/2	250	195	232	121	132	195	235	127	200	644	195	235	440	220	69
TBE80-19/2	250	215	249	121	132	195	235	127	200	670	195	235	440	220	75
TBE80-23/2	300	260	271	121	132	195	235	127	240	787	195	235	440	220	96
TBE80-29/2	300	260	271	121	132	195	235	127	240	787	195	235	440	220	101
TBE80-30/2	350	314	382	167	175	195	235	145	275	918	195	235	500	250	178
TBE80-38/2	350	314	382	167	175	195	235	145	275	918	195	235	500	250	193
TBE80-47/2	350	314	382	167	175	195	235	145	275	962	195	235	500	250	211

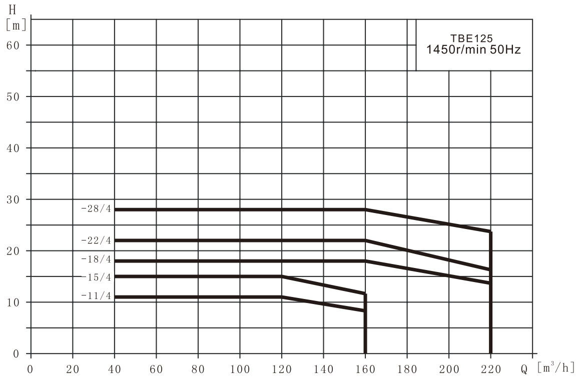
TBE100

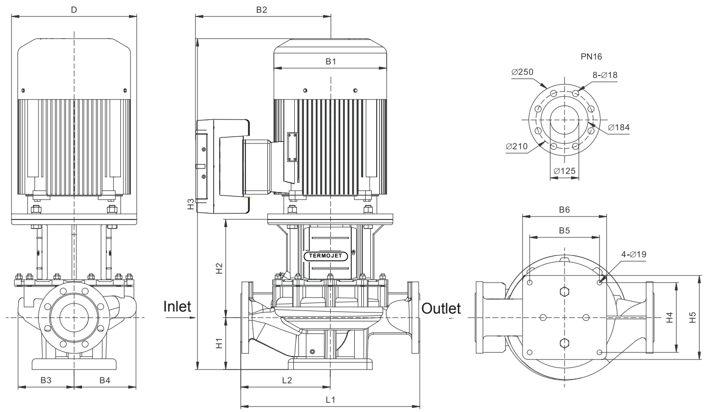




Model	Dimensions, mm															Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	D1	
TBE100-10/2	250	195	232	121	148	195	235	135	190	642	195	235	450	225	4-ø14	72
TBE100-15/2	250	215	249	121	148	195	235	135	190	668	195	235	450	225	4-ø14	78
TBE100-17/2	300	260	271	121	148	195	235	170	230	820	195	235	500	250	4-ø14	104
TBE100-22/2	300	260	271	121	148	195	235	170	230	820	195	235	500	250	4-ø14	110
TBE100-27/2	350	314	382	123	148	195	235	170	265	933	195	235	550	275	4-ø14	172
TBE100-34/2	350	314	382	123	148	195	235	170	265	933	195	235	550	275	4-ø14	187
TBE100-40/2	350	314	382	167	167	290	350	170	270	982	290	350	550	275	4-ø19	227

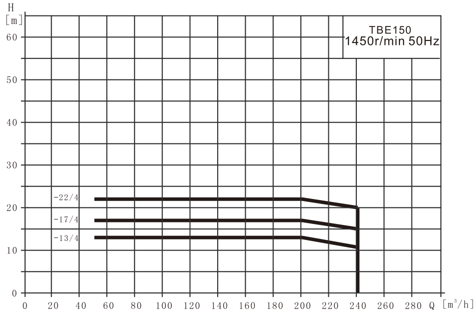
TBE125

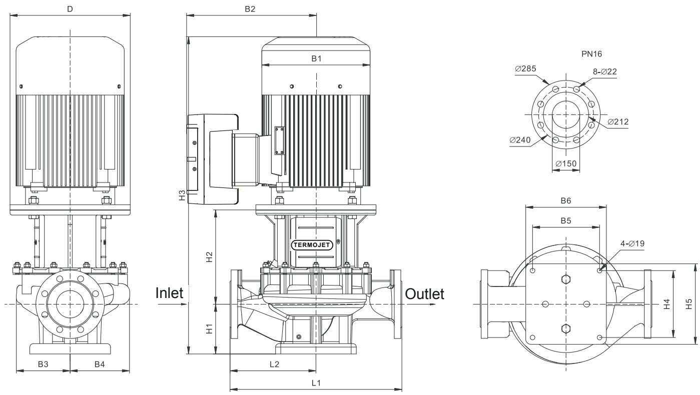




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE125-11/4	300	260	271	170	205	290	350	245	235	900	290	350	620	310	156
TBE125-15/4	300	260	271	170	205	290	350	245	235	900	290	350	620	310	165
TBE125-18/4	350	314	382	191	225	290	350	245	290	1033	290	350	800	400	254
TBE125-22/4	350	314	382	191	225	290	350	245	290	1077	290	350	800	400	270
TBE125-28/4	350	355	398	219	248	290	350	245	285	1108	290	350	800	400	318

TBE150

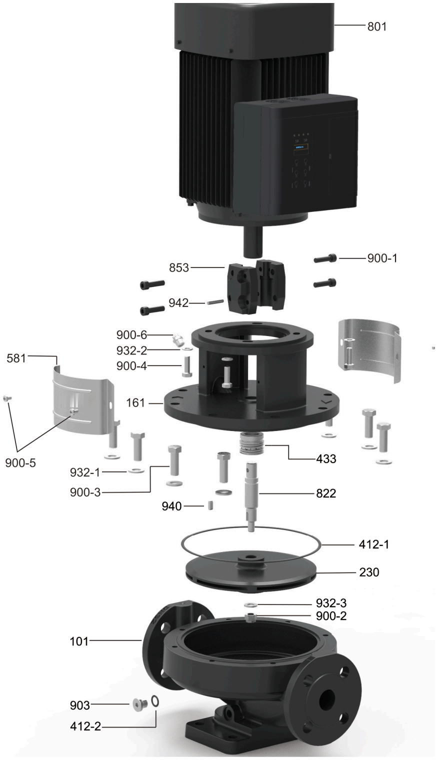




Model	Dimensions, mm														Weight, kg
	D	B1	B2	B3	B4	B5	B6	H1	H2	H3	H4	H5	L1	L2	
TBE150-13/4	350	314	382	202	242	290	350	245	275	1018	290	350	800	400	254
TBE150-17/4	350	314	382	202	242	290	350	245	275	1062	290	350	800	400	270
TBE150-22/4	350	355	398	231	265	290	350	245	285	1108	290	350	800	400	318

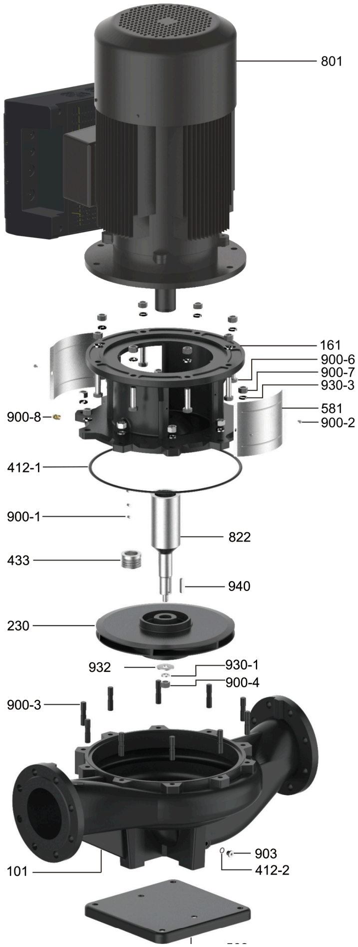
X. Exploded views of the pump

TBE32



No.	Part name	Material
101	Pump body	HT200
161	Pump cover	HT200
230	Impeller	HT200
412-1	O-ring seal	NBR
412-2	O-ring seal	FPM
433	Mechanical seal	/
581	Protective film	304
801	Electrical machinery	/
822	Pump shaft	304
853	Half coupling	F0212J
900-1	Hexagon socket head cap screw	Q235A
900-2	Locking nut	304
900-3	Hexagonal head bolt	304
900-4	Hexagonal head bolt	304
900-5	Cross recessed pan head screws	Q235A
900-6	Bleed cock	H59
903	Internal hexagonal plug	304
932-1	Flat washer	304
932-2	Flat washer	304
932-3	Flat washer	304
940	Flat key	45#
942	Cylindrical pin	304

TBE40 — TBE150



No.	Part name	Material
101	Pump body	HT200 / HT250
161	Pump cover	HT200 / HT250
230	Impeller	HT200 / Stainless steel
412-1	O-ring seal	NBR / FPM
412-2	O-ring seal	FPM
433	Mechanical seal	/
562	Base	Q235A
581	Protective film	304
801	Electrical machinery	/
822	Pump shaft	06Cr19Ni10 (304) + 45#
900-1	Hexagon socket set screw with concave end	Q235A
900-2	Cross recessed pan head screws	Q235A
900-3	Double headed studs	Q235A
900-4	Locking nut	304
900-5	Hexagonal head bolt	Q235A
900-6	Hexagonal head bolt	Q235A
900-7	Type 1 hex nut	Q235A
900-8	Bleed cock	H59
903	Internal hexagonal plug	304
930-1	Spring washer	304
930-3	Spring washer	65Mn
932	Impeller washer	304
940	Flat key	304

XI. Analysis of common faults

1. Common pump faults

WARNING
The power supply must be disconnected and secured against accidental switching on before disassembling the pump or adjusting the position of the junction box. Make sure that the liquid overflowing after the pump body is disassembled will not cause personal injury or damage other components; pay special attention to high-temperature, low-temperature, toxic and corrosive liquid. Promptly have the pump inspected and repaired by professional personnel to eliminate the problem if it malfunctions.

Fault	Cause analysis
Motor cannot start	1. Power failure 2. Fuse blown 3. Overload tripping of motor starter 4. Poor contact or coil failure of motor starter contacts 5. Fuse blown of control circuit 6. Motor failure
The pump can run but there is no water	1. The inlet is blocked or there are impurities in the pump chamber 2. Leakage in the inlet pipe or excessive leakage at the wear ring of pump cavity 3. The bottom valve or check valve is not open 4. There is air inside the inlet pipe or pump 5. Motor reversal 6. Wrong water pump is selected 7. The suction head of the water pump is too high
Occasional tripping of motor starter	1. Too low overload setting 2. Periodic fluctuations in supply voltage 3. The pressure difference between the inlet and outlet of the water pump is too low
The motor reverses after the power supply is disconnected	1. Leakage at the water inlet 2. Damaged bottom valve or check valve 3. The bottom valve or check valve is stuck in the fully open or partially open position
The operating current of the water pump is too high	1. Excessive flow 2. The motor bearing is damaged 3. Mechanical friction between impeller and pump chamber

Fault	Cause analysis
Water pump leakage	1. The pump shaft is not installed in the correct place 2. Damaged shaft seal 3. Damaged O-ring 4. Defects in water pump castings
The motor starter is connected but the pump is not started	1. Failure of power circuit 2. Fuse blown 3. Failure of motor starter and coil contact 4. Failure of control circuit
The motor starter immediately trips when the power switch is turned on	1. Circuit fault 2. Failure of motor starter 3. Loose cable connection 4. Motor winding is damaged 5. The pump is blocked and stuck by foreign matters 6. Too low overload setting
The flow rate of the pump is unstable	1. The inlet pressure of the water pump is too low 2. There are foreign objects blocking the inlet pipeline or pump chamber 3. There is air in the pipeline and pump
Noise of water pump	1. Water pump idling 2. Cavitation of water pump 3. Friction of parts in pump cavity 4. Resonance of the device 5. Foreign matters stuck in the pump 6. Unstable pipeline support 7. Air in liquid

2. Common faults of the frequency converter

2.1. No display with power on

- Check with a multimeter whether the input power of the frequency converter is the same as its rated voltage.
- Check whether the three-phase rectifier bridge is in good condition, and seek service if the rectifier bridge is damaged.

2.2. Tripping of the power supply air switch with power on

- Check whether there is an earth fault or a short circuit between the input power supply, and eliminate the problem.
- Check whether the rectifier bridge has broken down, and seek service if it has been damaged.

2.3. The motor does not rotate after the frequency converter runs

- Check whether there is a balanced three-phase output between U, V and W. If so, check whether the motor is damaged or locked, and confirm whether the motor parameters are set correctly.
- Seek service if there is output but the three phases are unbalanced.
- Seek service if there is no output voltage.

2.4. The converter displays normally, but the air switch jumps off after operation

- Check whether there is a short circuit between the output modules, and seek service if so.
- Check whether there is a short circuit or an earth fault between the motor leads, and eliminate it if so.
- An output AC reactor shall be considered if the tripping occurs occasionally and there is a long distance between the motor and the frequency converter.

2.5. Failure to shut down when the water draw-off has stopped

- Check whether the feedback pressure displayed on the frequency converter panel is greater than or equal to the set pressure. If the feedback is less than the set value, check whether the range of the pressure sensor is set correctly, whether it is reversed, whether there is air, and whether the water inlet is blocked by sundries.
- Manually shut down the frequency converter and observe whether the pressure drops. Replace the check valve if the feedback value changes near the set value.

2.6. Failure to sleep normally in case of a small amount of water or water leakage

- Reduce the leakage coefficient **F0.04** if the pump cannot sleep or sleeps for too long.
- Increase the leakage coefficient **F0.04** if the pump sleeps in advance and starts and stops frequently.

2.7. No protection shutdown in case of water shortage

- The water shortage protection switch **F4.00** is not opened.
- The allowable threshold **F4.01** for water shortage fault detection is set too low.
- The detection current percentage **F4.04** for water shortage protection is set too low.

XII. Fault code and handling method

Fault code	Fault type	Cause analysis	Countermeasures
E002	Acceleration operation over-current	1. Acceleration is too fast 2. Low grid voltage 3. The power of the frequency converter is too low	1. Increase acceleration time 2. Check the input power supply 3. Choose a frequency converter with higher power
E003	Deceleration operation over-current	1. Deceleration is too fast 2. The power of the frequency converter is too low	1. Increase deceleration time 2. Increase the power of the frequency converter
E004	Constant speed operation over-current	1. Sudden or abnormal load 2. Low grid voltage 3. The power of the frequency converter is too low	1. Check load or reduce sudden changes in load 2. Check the input power supply 3. Choose a frequency converter with higher power
E005	Acceleration operation overvoltage	1. Abnormal input voltage 2. Re-start the rotating motor after instantaneous power outage	1. Check the input power supply 2. Avoid stopping and re-starting
E006	Deceleration operation overvoltage	1. Deceleration is too fast 2. Large load inertia 3. Abnormal input voltage	1. Increase deceleration time 2. Increase energy consumption of braking components 3. Check the input power supply
E007	Constant speed operation overvoltage	1. Abnormal change in input voltage 2. Large load inertia	1. Install input reactor 2. Add appropriate energy consumption braking components
E008	Buffer resistor overload	1. Abnormal input voltage 2. Deceleration is too fast 3. Large load inertia	1. Check the input power supply 2. Increase deceleration time 3. Increase energy consumption of braking components
E009	Bus under-voltage	Low grid voltage	Check the input power supply of the power grid
E010	Frequency converter overload over-current	1. Acceleration is too fast 2. Re-start the rotating motor 3. Low grid voltage 4. Excessive load	1. Increase acceleration time 2. Avoid stopping and re-starting 3. Check the grid voltage 4. Choose a frequency converter with higher power
E011	Motor overload	1. Low grid voltage 2. The rated current of motor is not set correctly 3. The motor is locked or the load is largely changed 4. High-power motor drives light load	1. Check the grid voltage 2. Reset the rated current of the motor 3. Check the load and adjust the increase amount of torque 4. Choose the appropriate motor

Fault code	Fault type	Cause analysis	Countermeasures
E012	Phase loss on the input side	Phase loss in input R, S, T	1. Check the input power supply 2. Check the installation wiring
E013	Phase loss on the output side	U, V, W phase-failure output (or severe three-phase asymmetry of the load)	Check the motor cable and the motor windings
E014	Module overheating	1. Instantaneous over-current of frequency converter 2. There is a phase-to-phase or ground short circuit in the output three-phase 3. Blocked air duct or damaged fan 4. The ambient temperature is too high 5. Loose control board wiring or plugins 6. Failure of power circuit 7. Abnormal board making	1. Refer to over-current countermeasures 2. Re-wiring 3. Dredge the air duct or replace the fan 4. Reduce ambient temperature 5. Check and re-connect 6. Seek services
E015	External fault	External fault input terminal action	Check external device input
E016	Communication fault	1. Improper Baud setting 2. Communication error in serial communication 3. Long communication interruption	1. Set appropriate Baud 2. Press the RUN/STOP key to reset and seek service 3. Check the communication interface wiring
E017	Power-on relay fault	The relay is not engaged	Replace the power-on relay or seek technical support
E018	Circuit fault through current detection	1. Poor contact of the control board connector 2. Failure of power circuit 3. Hall device damage 4. Abnormal amplification circuit	1. Check the connector and plug it in again 2. Seek services
E021	Data overflow	1. Error in reading and writing control parameters 2. EEPROM is damaged	1. Replace the control board 2. Seek services
E022	Error in parameter verification	Check whether any parameters have been modified	Press the RUN/STOP key to reset

Fault code	Fault type	Cause analysis	Countermeasures
E023	Short circuit to ground fault	Check whether the motor is short circuited to ground	1. Replace the cable or motor 2. Seek services
E024	Feedback disconnection fault	1. The sensor is disconnected or in poor contact 2. The detection time of disconnection is too short 3. The sensor is damaged or the system has no feedback signal	1. Check installation and wiring of sensor 2. Adjust the detection time of wire breakage 3. Replace the sensor
E025	Power-on time up	Power-on time to set time	1. Press the RUN/STOP key to reset 2. Seek services
E026	Run time up	Run time to set time	1. Press the RUN/STOP key to reset 2. Seek services
E027	Water shortage alarm	1. Abnormal water pressure/level 2. The sensor is disconnected or in poor contact, and the system has no feedback signal 3. The detection time of water shortage protection is too short 4. The detection frequency of water shortage protection is too low 5. The detection current of water shortage protection is too high	1. Check if the water pressure at the water pump inlet is abnormal 2. Check installation and wiring of sensor 3. Check the relevant parameter settings
E028	High water pressure alarm	1. Abnormal sensor feedback signal 2. The adjustment of the high-pressure alarm value is too low	1. Detect sensor wiring 2. Check the relevant parameter settings
E029	Low water pressure alarm	1. The low-pressure alarm value is set too high 2. The sensor is disconnected or in poor contact, and the system has no feedback signal 3. The sensor type is not selected according to the actual situation	1. Modify parameters 2. Detect sensor
E031	Pipe burst alarm	The detection time for pipe burst is too short	Check the pipeline
E040	Wave-by-wave current limiting fault	1. Whether the load is too large or the motor is blocked 2. The power of the frequency converter is too low	1. Reduce the load and check the motor and mechanical condition 2. Choose a frequency converter with higher power
E050	Error in multiple online communication	1. Abnormal communication in multiple online systems 2. Duplicate networking addresses in multiple online systems	1. Power on again 2. Check the setting of CAN network communication address 3. Seek services

XIII. Parameter table of the frequency converter

Symbols in the «Change» column

- ○ the set value of the parameter can be changed when the frequency converter is in standby or running mode;
- ● the set value of the parameter cannot be changed when the frequency converter is in operation;
- ◎ the value of the parameter is the actual detection value and cannot be changed.

1. Parameters displayed on the panel

Press the **SHIFT** key to switch the display.

Running state

Display	Name	Note	Unit	Change
P	Current pressure	Real-time pressure of the system	Bar	◎
H	Operating frequency	Current operating frequency	Hz	◎
d	Set pressure	Set pressure of system	Bar	◎
A	Operating current	Output current of machine	A	◎
U	Bus voltage	Bus voltage of machine	V	◎

Shutdown state

Display	Name	Note	Unit	Change
P	Current pressure	Real-time pressure of the system	Bar	◎
d	Set pressure	Set pressure of system	Bar	◎
U	Bus voltage	Bus voltage of machine	V	◎

2. Common parameter group

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F0.00	Pressure setpoint	F4.01 ~ F0.10	0.1 bar	3.0 bar	○	In a multi-pump network the pressure is set on the master only.
F0.01	Start deviation	0,0 ~ F0.00	0.1 bar	0.3 bar	○	The pump wakes from sleep mode when the pressure falls below the setpoint by this deviation.
F0.02	Direction of motor rotation	0: forward 1: reverse	1	0	●	Reverses the direction in which the pump runs.
F0.03	Frost protection	0: off 1: on, in seconds 2: on, in minutes	1	0	○	The detailed settings for frost and seizing protection are F4.10–F4.12. In a multi-pump network the anti-frost measures are set separately.
F0.04	Leakage coefficient	0.0 s ~ 100.0 s	0.1 s	5.0 s	○	The greater the leakage, the smaller the coefficient.
F0.05	Source of the start and stop signal	0: from the keypad 1: from the terminals 2: over communication	1	0	○	In a multi-pump network the slaves are set to 2.
F0.06	Automatic start	0 ~ 1	1	0	○	0: off; 1: on.
F0.07	Self-start delay	0.0 s ~ 100.0 s	s	5.0 s	○	The delay before the automatic start.
F0.08	Sensor range	0.0 bar ~ 200.0 bar	0.1 bar	16.0 bar	○	The maximum range of the sensor concerned.
F0.09	Sensor feedback channel	0: AI1 1: AI2 2: max (AI1, AI2) 3: min (AI1, AI2)	1	2	○	AI1 — current feedback by default, AI2 — voltage feedback.

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F0.10	High pressure alarm setpoint	F0.00 ~ F0.08	0.1 bar	14.4 bar	○	The unit alarms and stops 0.1 s after the feedback pressure reaches the setpoint. The alarm resets automatically after the reset delay once the pressure returns to normal.
F0.11	Low pressure alarm setpoint	0,0 ~ F0.00	0.1 bar	0.0 bar	○	The unit alarms and stops after the F4.09 delay when the feedback pressure is below the setpoint. It resets automatically once the pressure returns to normal; a value of 0 disables the function.
F0.12	Inlet stop pressure	0.0 bar ~ F0.08	Bar	3.5 bar	○	Valid when F0.05 = 3: the converter starts when the inlet pressure is below F0.13 and stops when the pressure is above F0.12. The inlet pressure sensor may be AI1 or AI2.
F0.13	Inlet start pressure	0.0 bar ~ F0.12	Bar	2.5 bar	○	
F0.14	Inlet feedback channel	0: AI1 1: AI2	0	0	○	
F0.15	Converter operating mode	0 ~ 1	1	0	○	0: constant pressure control; 1: ordinary speed control.
F0.16	Product identifier	/	1	/	◎	Factory setting.
F0.17	Software version	2.000 ~ 2.999	1	—	◎	This manual applies to this range of versions only.
F0.18	Acceleration time	0.0 s ~ 6500.0 s	0.1 s	5.0 s	○	Depends on the rating.
F0.19	Deceleration time	0.0 s ~ 6500.0 s	0.1 s	5.0 s	○	Depends on the rating.

3. Protection parameter group of the water pump

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F4.00	Dry-run protection function	0 ~ 2	1	2	○	0: off; 1: by frequency and current; 2: by outlet pressure; 3: by frequency, current and pressure; 4: by inlet pressure.
F4.01	Dry-run detection threshold	0.0 bar ~ F0.00	0.1 bar	0.5 bar	○	Dry running is detected when the feedback pressure is below the setpoint.
F4.02	Dry-run detection frequency	0.00 Hz ~ upper frequency limit F2.07	0.00 Hz	48.00 Hz	○	The comparison frequency: dry running is detected when the operating frequency is above it.
F4.03	Dry-run detection time	0.0 s ~ 200.0 s	0.1 s	60.0 s	○	The dry-run alarm is raised after the conditions have been met for this time.
F4.04	Dry-run detection current, percent	0,0 % ~ 100,0 %	0,1 %	40,0 %	○	Valid when F4.00 = 1. A percentage of the rated motor current: dry running is detected when the operating current is below this value.
F4.05	Automatic restart delay after dry running	0 min ~ 9999 min	1 min	15 min	○	0 — the dry-run alarm is reset through F4.07 and F4.08; other than 0 — the set delay is counted after dry running is detected.
F4.06	Number of automatic resets of the dry-run protection	0 ~ 9999	1	10	○	After an alarm the converter resets automatically and starts again after the F4.05 time. The number of resets is limited by this parameter; once it is used up, the alarm is reset by hand with the RUN/STOP key. 9999 — an unlimited number of resets.

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F4.07	Water arrival detection pressure	0.0 bar ~ F0.00	0.1 bar	1.0 bar	○	The E027 alarm is cleared automatically if the inlet water pressure is greater than or equal to the set value and holds for longer than the detection time. Suitable for systems with a positive inlet head; the pressure value is the outlet pressure.
F4.08	Water arrival detection time	0.0 s ~ 100.0 s	0.1 s	20.0 s	○	
F4.09	Abnormal pressure alarm delay	0.0 s ~ 120.0 s	0.1 s	3.0 s	○	The delay of the water pressure alarm and of the alarm signal.
F4.10	Anti-frost operating frequency	0.00 Hz ~ upper frequency limit F2.07	Hz	10.00 Hz	○	The time unit of the frost and seizing protection is seconds or minutes, depending on F0.03. If the interval is 0, the pump runs continuously at the anti-frost frequency.
F4.11	Anti-frost run duration	0 ~ 65000 s/min	s/min	60 s/min	○	
F4.12	Anti-frost run interval	0 ~ 65000 s/min	s/min	300 s/min	○	
F4.13	Pipe burst detection time	0 s ~ 1000 s	1 s	0 s	○	The E030 alarm is raised after the F4.10 time when the operating frequency of every converter in the system is greater than or equal to F4.02 and the pressure is below the start pressure. A value of 0 disables the detection.

4. Protection and fault parameter group

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F6.00	Motor overload protection	0: disabled 1: enabled	1	1	○	0: no protection; 1: with protection.
F6.01	Motor overload protection coefficient	0,20 ~ 10,00	0,01	1,00	○	
F6.02	Self-recovery interval after a fault	0.1 s ~ 100.0 s	s	30.0 s	○	
F6.03	PWM settings	Units — PWM mode: 0: automatic switching; 1: CPWM; 2: DPWM; 3: SPWM. Tens — voltage prediction compensation. Hundreds — 0: SSSU, 1: DSDU. Thousands — random carrier: 0: random carrier; 1: random zero vector		0x0111	○	
F6.04	Hardware current and voltage protection	Units — hardware current limiting (CBC): 0: off, 1: on. Tens — hardware overvoltage protection: 0: off, 1: on. Hundreds — SC filtering time 0–F (0 — SC protection off). Thousands — current noise suppression: 0: off, 1: on.		1111	○	SC filtering time: 1.2 μs × T, where T is set by the hundreds digit.

Function code	Description of function code	Set range	Minimum unit	Factory value	Change	Remarks
F6.05	CBC protection point	0 ~ 220 %	1 %	200 %	○	The adjustable CBC point as a percentage of the rated converter current.
F6.06	CBC overload protection time	1 ~ 5000 ms	ms	500 ms	○	
F6.07	Earth fault detection at power-on	0: off 1: on	1	1	○	
F6.08	Phase loss protection	Units — output phase loss protection. Tens — input phase loss protection. Hundreds — motor overload protection. 0: off; 1: on		0x0011	○	
F6.09	Software phase loss protection threshold	0,0 % ~ 999,9 %	0,1 %	20,0 %	○	The ratio of the input phase loss detected in software to the bus voltage.
F6.10	Output power correction coefficient	0 ~ 1000 %	1 %	100 %	○	Correction of the C-10 output power value.
F6.11	Bus undervoltage protection point	40,0 % ~ 150,0 %	0,1 %	100,0 %	○	Relative to 537 V.

XIV. Main circuit terminals and functions

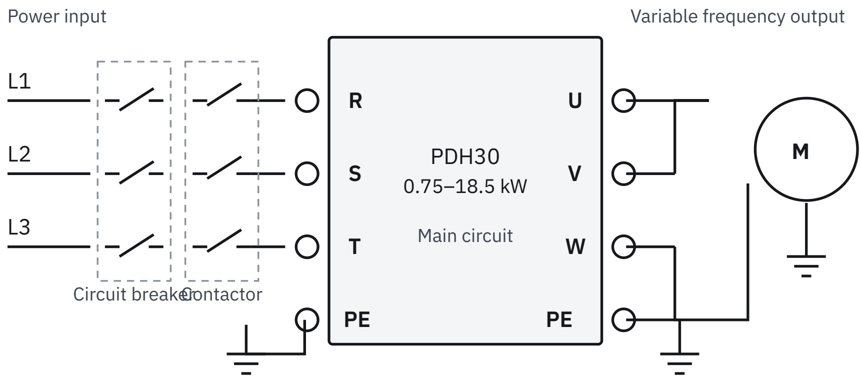


Figure 10. Schematic diagram of the main terminals of the 0.75–18.5 kW converter

Terminal marker	Name	Note
R, S, T	Three-phase power input terminal	Three-phase AC power input connection terminal
U, V, W	Frequency converter output terminal	Connect three-phase motor
PE	Grounding terminal	Connect grounding terminal

The graphics in this manual are schematic diagrams. The product is constantly being updated, so the pump purchased (including but not limited to appearance and colour) may differ from the one shown: be guided by the actual product.

Website	termojet.com.ua
E-mail	termojet@sofievka.kiev.ua
Phone	+380 50 718 91 65 +380 50 450 64 24
Address	Sofiivska Borshchahivka, 3 Kyivska St., Ukraine